Bivariate Normal Properties

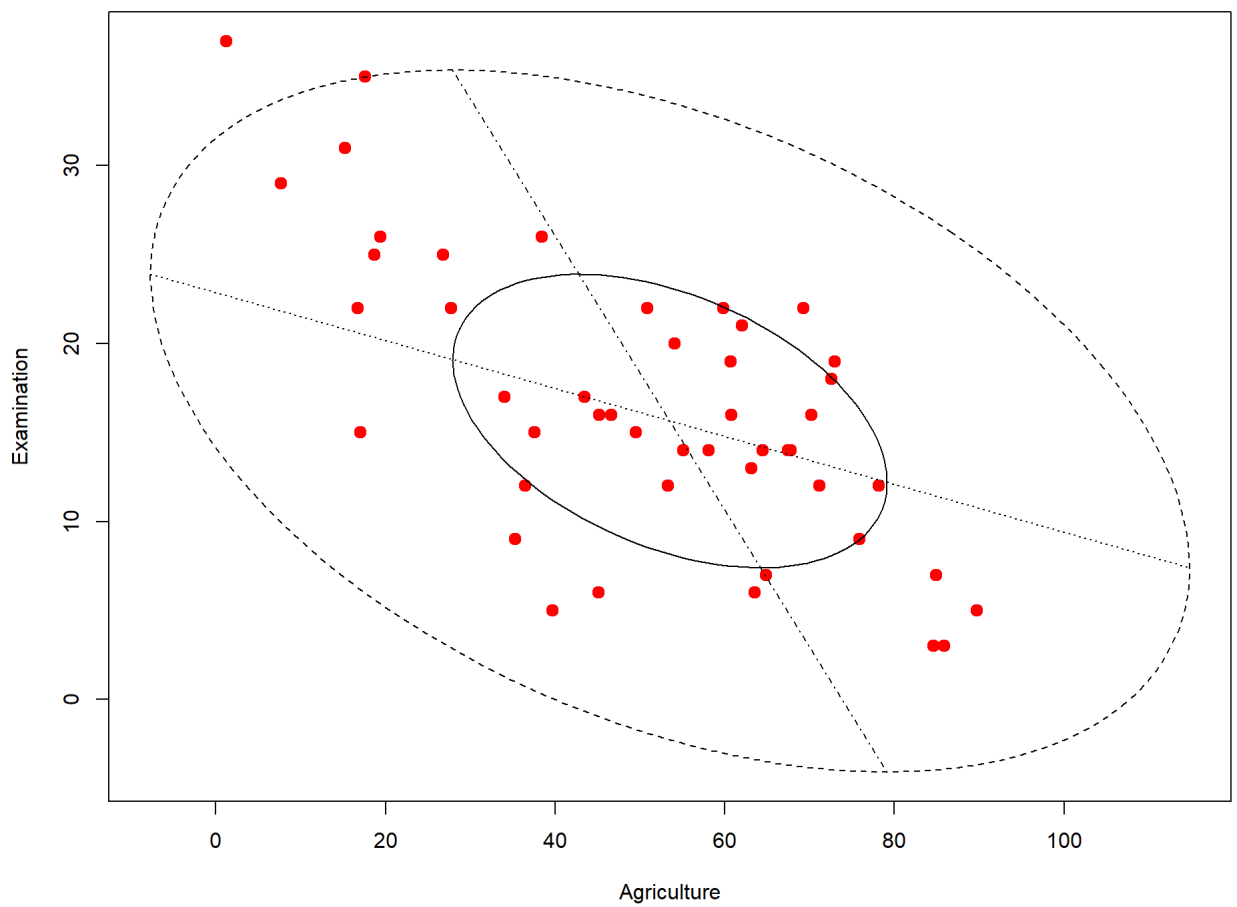
Dr. Jyotishka Datta

Bivariate boxplot

Instead of the central box containing 50% of the univariate data as in a univariate boxplot, the bivariate boxplot includes a pair of estimated ellipses, the inner one containing approximately 50% of the data and the outer one containing about 95%.

R code

```
library(MVA) # library for bvbox
library(datasets)
swiss2 <- swiss[, c("Agriculture", "Examination")]
bvbox(swiss2, xlab = "Agriculture", ylab = "Examination",pch = 19, cex = 1.25, col = "red")
```

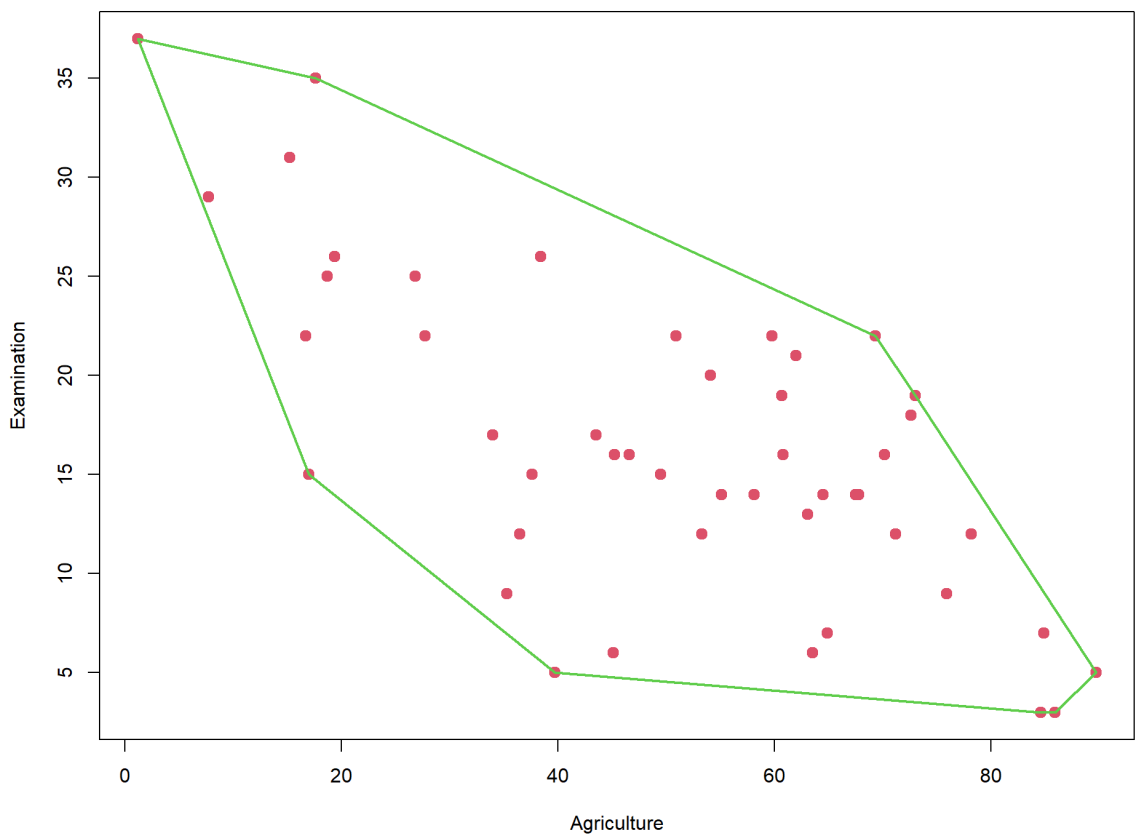


Convex Hull

The convex hull is a lot like wrapping an irregularly shaped object with the smallest amount of gift paper: The most extreme points in any direction will define the ultimate shape of the package. Mathematically, the convex hull is the smallest convex polygon containing all of the data.

R code

```
ch <- chull(swiss2) # find the indices of the convex hull
ch <- c(ch,ch[1]) # loop back to the beginning
plot(swiss2, pch = 19, col = 2,
     cex = 1.25) # plot the original data
lines(swiss2$Agriculture[ch], swiss2$Examination[ch],
      type = "l", col = 3, lwd = 2) # bold, green lines
```

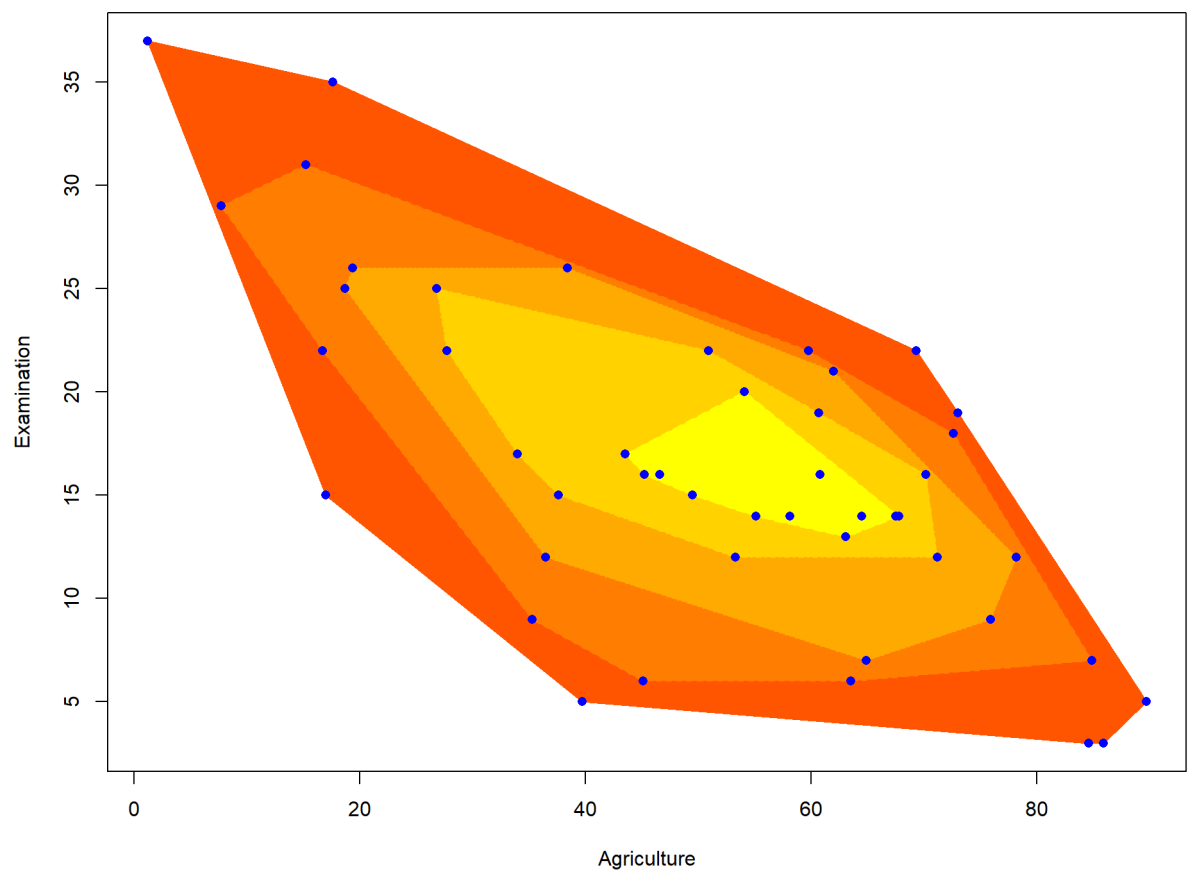


Layers

Suppose we omitted the observations on the convex hull and looked at what remains. This would remove the most extreme values to reveal those less extreme. We might then draw the convex hull of those remaining observations. It is also useful to repeat this process several times. The net effect is much like peeling an onion, successively removing the outermost layers, one at a time. This repeated process will give a better idea about the underlying shape of the data.

R code

```
library(aplpack)
nlev <- 5 # Number of levels
colors <- heat.colors(9)[3:(nlev+2)]
plothulls(swiss2, n.hull = nlev, col.hull = colors, xlab = "Agriculture", ylab = "Examination",lty.hull
points(swiss2, pch = 16, cex = 1, col = "blue")
```



Bivariate normal

The probability density function of a bivariate normal distribution with means μ_1, μ_2, standard deviations σ_1, σ_2, and correlation coefficient ρ is given by:

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}$$
$$\exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_x)^2}{\sigma_x^2} - 2\rho\frac{(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} + \frac{(y-\mu_y)^2}{\sigma_y^2}\right.\right.$$

Simplify

- If you standardize both x and y, i.e.,

$$x \mapsto (x - \mu_x)/\sigma_x, y \mapsto (y - \mu_y)/\sigma_y$$

.

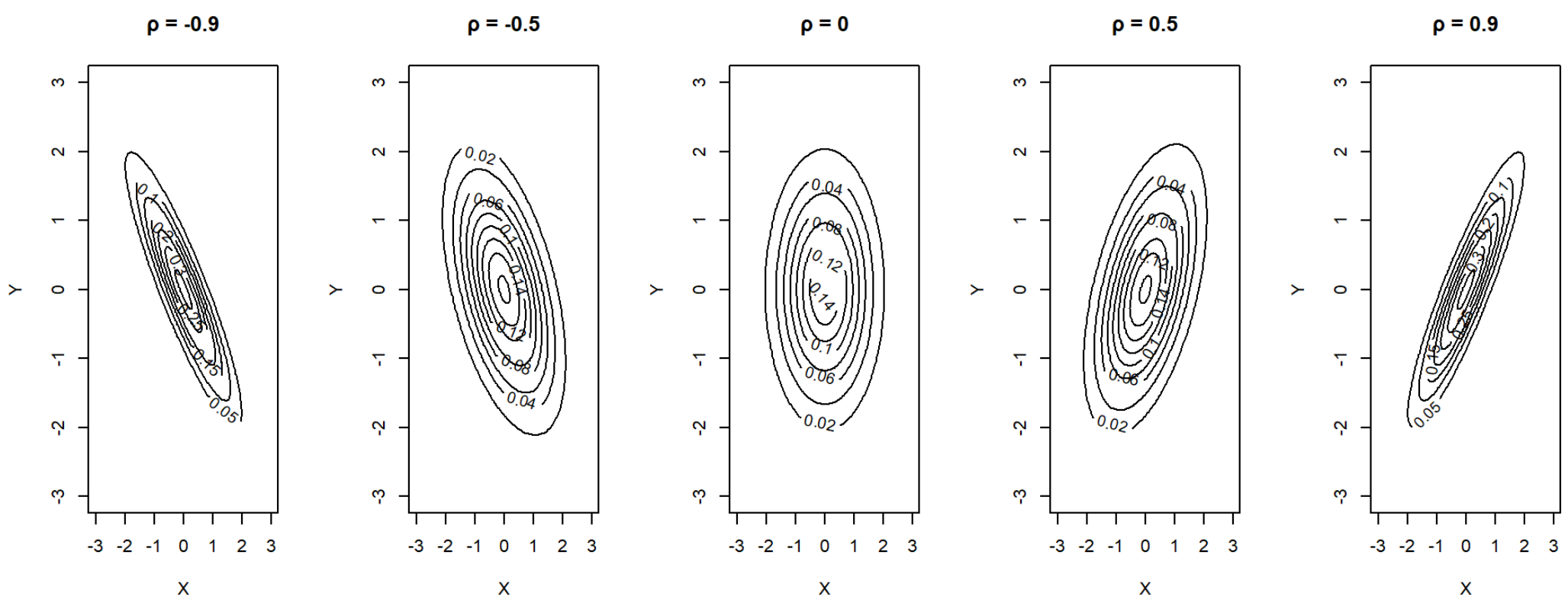
- We can get a much simpler form:

$$f(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[x^2 - 2\rho xy + y^2\right]\right)$$

- Clearly, if $\rho = 0$, this is just the product of two standard normal density functions.

Bivariate contours

- The R functions for random variable generation, CDF and quantile are respectively, `rmvnorm`, `dmvnorm`, `pmvnorm` and `qmvnorm` from the `mvtnorm` library.



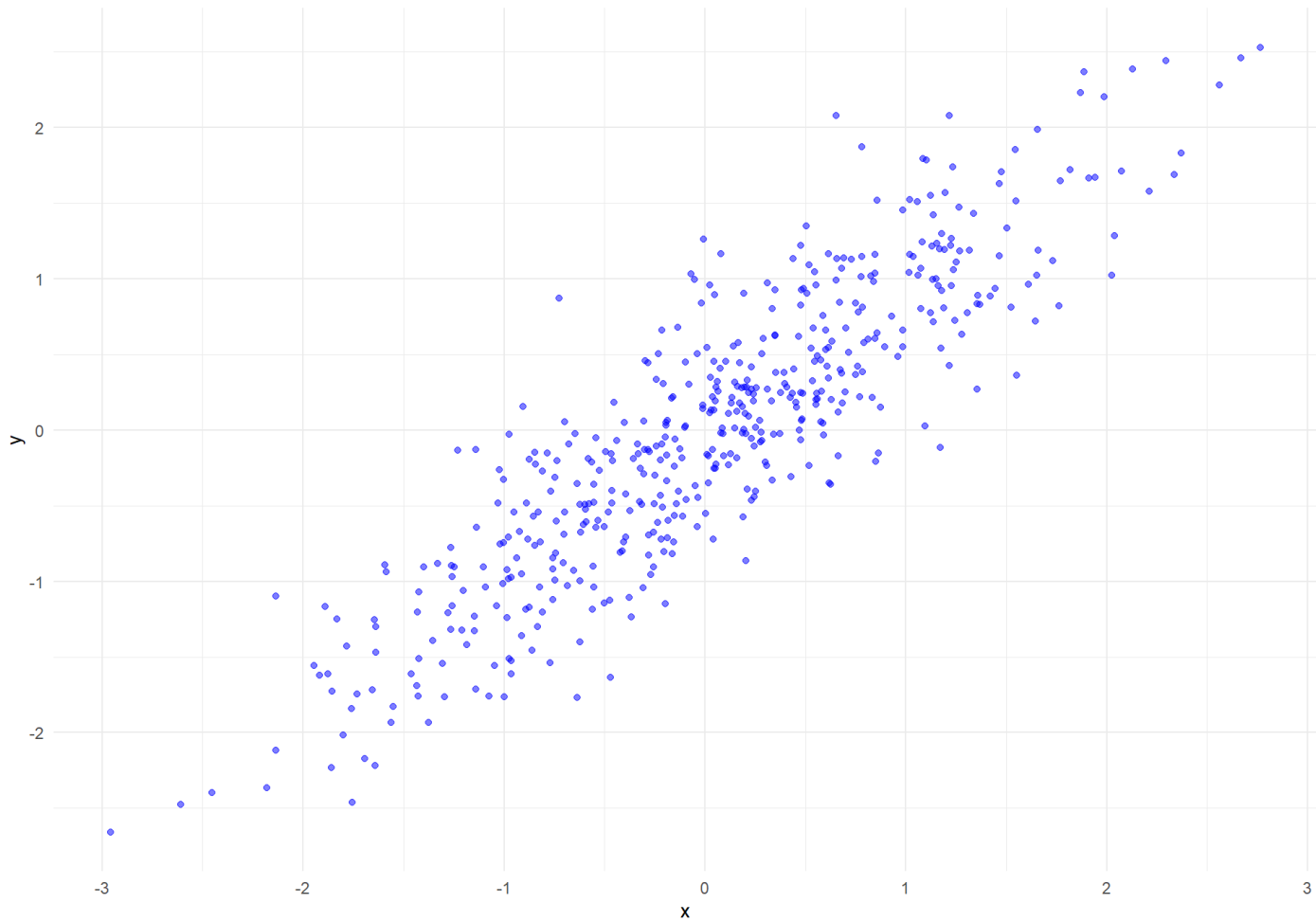
More ...

```r
library(MASS)  # For mvrnorm()
# Define parameters
set.seed(123)  # For reproducibility
n <- 500  # Number of samples
rho <- 0.9  # Correlation
mu <- c(0, 0)  # Mean vector
sigma <- matrix(c(1, rho, rho, 1), nrow = 2)  # Covariance matrix

# Generate samples using rmvnorm()
samples <- rmvnorm(n, mean = mu, sigma = sigma)
```

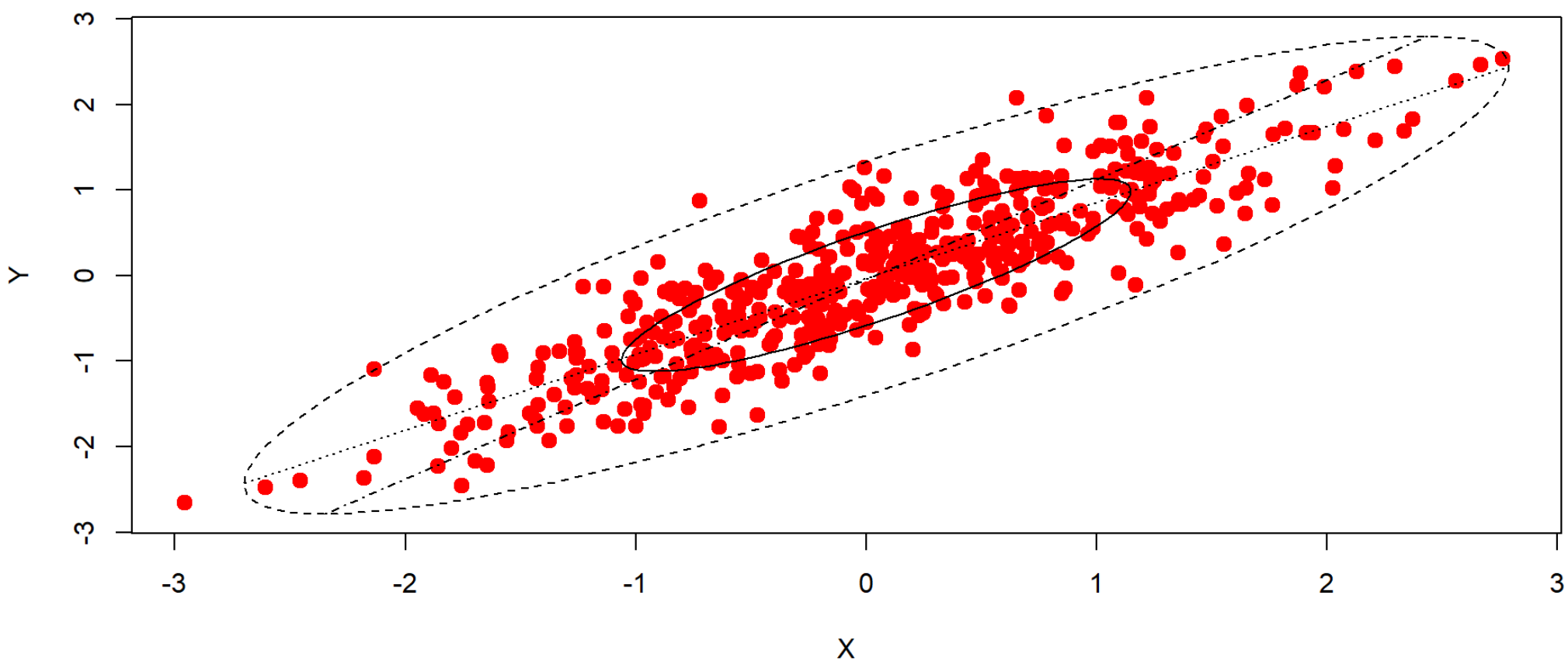
Plot them

```
library(ggplot2)
df <- data.frame(x = samples[,1], y = samples[,2])
(p <- ggplot(df, aes(x = x, y = y)) +
  geom_point(alpha = 0.5, color = "blue")+
  theme_minimal())
```



Bivariate boxplot

```r
bvbox(samples,pch = 19, cex = 1.25, col = "red")
```



Linear combinations

- A very useful property is that if X and Y are jointly normally distributed, any linear combination $\alpha X + \beta Y$ should be normally distributed.
- We use this property extensively in linear models as well as factor analysis.
- We can use this property to transform a correlated bivariate normal pair into two independent normals.

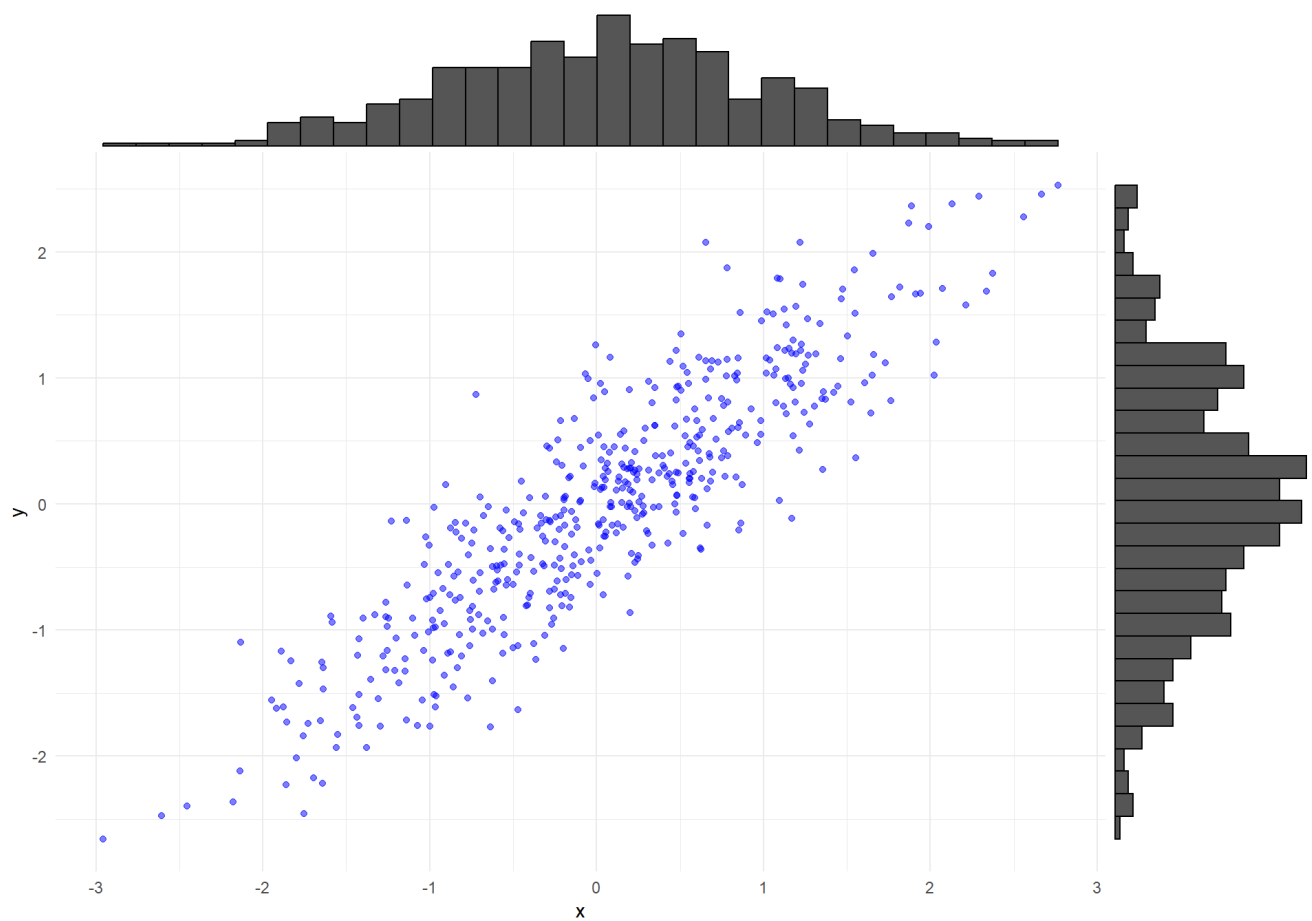
$$Z_1 = X/\sqrt{2}\sigma_X + Y/\sqrt{2}\sigma_Y, \quad Z_2 = X/\sqrt{2}\sigma_X - Y/\sqrt{2}\sigma_Y,$$

- Try to prove: (Z_1, Z_2) should be a bivariate normal with correlation zero.
- Try to work out the covariance of Z_1, Z_2 assuming unit variance first.

Marginal distributions

- If (X, Y) are jointly normal, then X and Y should have marginal distributions as normal.

```
library(ggExtra)
# with marginal histogram
(p1 <- ggMarginal(p, type="histogram"))
```

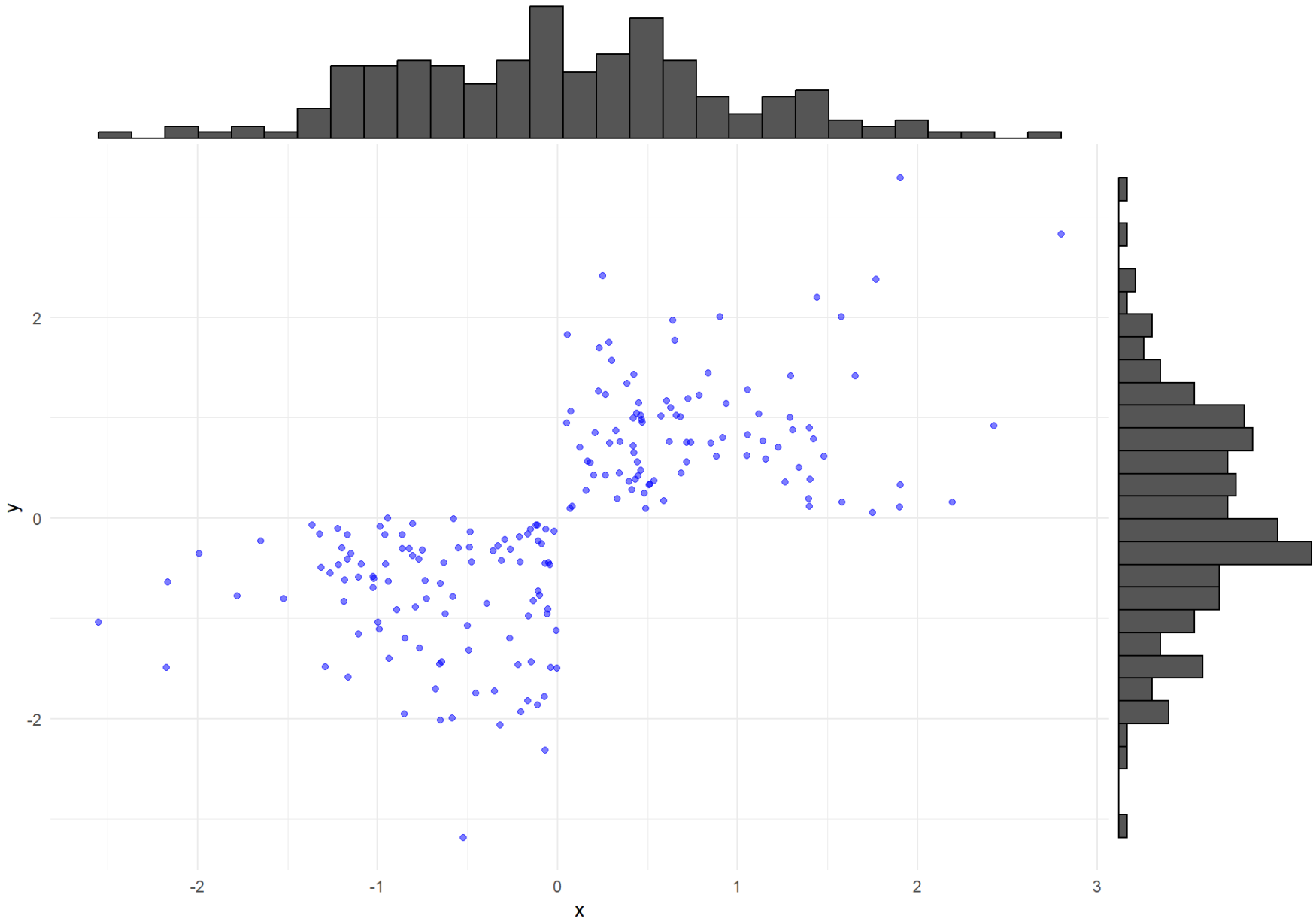


Is the reverse true?

- Jointly normal $\Rightarrow$ marginal normal, but marginally normal does not necessarily imply joint normal.

```
checker <- function(n)
# not bivariate normal
{
checker <- NULL # start a list
for (i in 1:n)
{
x <- rnorm(2)
if(x[1]>0) x[2] <- abs(x[2])
else x[2] <- -abs(x[2])
checker <- rbind(checker, x)
}
return(checker)
}
```

Look at the joint



Test for correlation

- We have already seen how to calculate the bivariate mean, variances, the dispersion matrix and correlation matrix.
- We can use `cor.test` for testing if the correlation is equal to zero.

```
cor.test(swiss$Agriculture, swiss$Examination)
```

```
    Pearson's product-moment correlation

data:  swiss$Agriculture and swiss$Examination
t = -6.3341, df = 45, p-value = 9.952e-08
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8133545 -0.4974484
sample estimates:
       cor
-0.6865422
```

Conditional distribution

- A useful property of bivariate normal is that the conditional distributions are also normally distributed.
- Specifically, for any given value or range of X, the corresponding values of Y are also normally distributed.
- In practical terms of data analysis, this suggests we can examine the Y values as normals, separately for different ranges of X.

Conditional

- When we sample from the bivariate normal density function, the conditional distribution of Y for any given value of X = x is also normally distributed with mean and variance:

$$\text{mean} = \mu_Y + \rho(x - \mu_X)\frac{\sigma_Y}{\sigma_X}, \text{ and variance} = (1 - \rho^2)\sigma_Y^2.$$

- The conditional mean of Y is a linear function of a known value of X.
- This property is utilized in linear regression.
- Also, the conditional variance of Y is always smaller than the marginal variance σ_Y^2.
- This reduction in the variance demonstrated the benefit achieved by knowing the value of X.

Conditional variance

$$\text{mean} = \mu_Y + \rho(x - \mu_X)\frac{\sigma_Y}{\sigma_X}, \text{ and variance} = (1 - \rho^2)\sigma_Y^2.$$

- Further, the conditional variance is the same for every value of X.
- One assumption of linear regression is the constant variance, conditional on X, yet independent of the value of X.
- This is called homoscedasticity.
- Let us connect this to simple linear regression.

Formulation

- Simple linear model: $Y = \beta_0 + \beta_1 X + \epsilon$.
- β_0 and β_1 are known as the intercept and slope, and ϵ is a random variable with mean zero and independent from X.
- Another way of looking at this is:

$$\mathbb{E}(Y|X = x) = \beta_0 + \beta_1 X, \text{ since } \mathbb{E}(\epsilon|x) = 0.$$

- LHS: conditional expectation of Y given X : how the mean of the random variable Y is changing according to a particular value, denoted by x, of the random variable X.
- we are saying is that the mean of Y is changing in a linear fashion with respect to the value of X.

Estimating β

- One way of estimating (β_0, β_1) is by minimizing the residual sum of squares:

$$RSS(\beta_0, \beta_1) = \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i)^2$$

- Look for estimates such that:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{(\beta_0, \beta_1) \in \mathbb{R}^2} RSS(\beta_0, \beta_1)$$

Estimating β

- It can be seen that the minimizers of the RSS are:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}, \hat{\beta}_1 = \frac{s_{xy}}{s_x^2}$$

- $\bar{X}$, s_{xy}, s_x^2: sample mean of Y, covariance between X and Y and variance of X (s_x is the std. dev.)
- Fitted values: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$. These are the vertical projections of $Y_1, \ldots, Y_n$ into the fitted line.

Now, compare

- Look at the fitted values for regressing Y on X.

$$\hat{Y}_i = \bar{Y} - \hat{\beta}_1 \bar{X} + \hat{\beta}_1 X_i = \bar{Y} - \frac{s_{xy}}{s_x^2}(X_i - \bar{X}) = \bar{Y} - \frac{r_{xy} s_y}{s_x}(X_i - \bar{X})$$

- and, look at the conditional mean/expectation of Y given X for $(Y, X) \sim BVN(\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho)$.

$$\text{mean} = \mu_Y + \rho(x - \mu_X)\frac{\sigma_Y}{\sigma_X}, \text{ and variance} = (1 - \rho^2)\sigma_Y^2.$$

- Then, convince yourself that they are essentially the same expression (albeit at the population level and at the sample level.)

BVN to MVN

- It is often convenient to express the variances and covariance in a symmetric matrix, denoted by Σ.
- which can be written in terms of the variances and correlation as:

$$\Sigma = \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix}$$

MVN

There is also a matrix representation of the above. Let us define $\boldsymbol{\mu} = (\mu_1, \mu_2)$ to be the vector of means and Σ in the last slide to denote the matrix of variances and covariance.

$$\det(\boldsymbol{\Sigma}) = \sigma_X^2 \sigma_Y^2 (1 - \rho^2)$$

In matrix notation

- The density function of the bivariate normal distribution can be written as

$$\phi(x \mid \mu, \Sigma) = (2\pi)^{-1} \det(\Sigma)^{-1/2} \exp\left\{ -\frac{1}{2}(x - \mu)' \Sigma^{-1} (x - \mu) \right\}$$

- **This** holds true for not just $p = 2$ but for any p, where the Σ is a $p \times p$ matrix and μ is a $p \times 1$ vector. This is the density for multivariate normal.